

Human Simulacra: A Step toward the Personification of Large Language Models

Qiujie Xie
Fudan University

Qiming Feng^{*}
Fudan University

Tianqi Zhang^{*}
Tongji University

Qingqiu Li
Fudan University

Yuejie Zhang[✉]
Fudan University

Rui Feng[✉]
Fudan University

Shang Gao
Deakin University

Abstract

Large language models (LLMs) are recognized as systems that closely mimic aspects of human intelligence. This capability has attracted attention from the social science community, who see the potential in leveraging LLMs to replace human participants in experiments, thereby reducing research costs and complexity. In this paper, we introduce a framework for large language models personification, including a strategy for constructing virtual characters' life stories from the ground up, a Multi-Agent Cognitive Mechanism capable of simulating human cognitive processes, and a psychology-guided evaluation method to assess human simulations from both self and observational perspectives. Experimental results demonstrate that our constructed simulacra can produce personified responses that align with their target characters. Our work is a preliminary exploration which offers great potential in practical applications. All the code and datasets will be released, with the hope of inspiring further investigations.

1 Introduction

Researchers in psychology and sociology have long relied on human participants to conduct experiments that explore patterns of human behaviors and mental states (Camerer et al., 2018; Folke et al., 2016; Qiu et al., 2017). However, this method often faces numerous challenges, including the difficulty of recruiting human participants, high costs, and ethical considerations. Against this backdrop, the potential of large language models (LLMs) to mimic human interactions is drawing increasing attention. Our question is: **Can LLMs simulate human cognitive processes, thereby assisting researchers in exploring human interactions without involving real human participants?**

There is already a discussion in the community, with psychologists and sociologists advocating the use of LLMs to replace human participants in experiments, thus reducing costs and complexity, and

providing more controllable experimental conditions in certain contexts (Dillion et al., 2023; Hutson, 2023; Grossmann et al., 2023; Li et al., 2023; Kjell et al., 2023). Despite the progress, the human simulations they created can only be used for group studies and cannot accurately replicate the nuanced responses of an individual. These simulations are hindered by limitations such as inconsistent performance across different types of tasks and a lack of depth in capturing the complex, context-driven characteristic of real human behaviors. The quest is on for a more advanced simulacrum that can produce consistent behaviors indistinguishable from those of humans.

Computer science researchers are also striving to humanize LLMs, with a particular emphasis on role-playing tasks. They focus on replicating the professional skills (Salewski et al., 2023) or conversational styles (Shao et al., 2023; Tu et al., 2023; Wang et al., 2023b) of target personas. However, as the goal of role-playing tasks is to convincingly portray the outward characteristics of target personas, these studies merely simulate the external traits of roles based on fragmented information about the character (e.g., brief introduction or scattered life experiences), lacking a holistic emulation of the inner patterns (e.g., emotion) of humans. Their models are unable to dynamically adjust the simulacrum's behavior or personality in response to interactions with humans, which limits the potential of these models to mimic a broader spectrum of human behaviors. There is a need to explore more generalized human simulations to meet the demands of practical applications, especially psychological and sociological experiments.

In light of the above concerns, in this paper, we define the task of LLM personification as shaping the intrinsic traits of LLM based on the complete life story of the target character, enabling the LLM to simulate their specific personality and interact with the external world in a human-like manner.

Given the definition, we propose a novel framework that includes a virtual character dataset, a multi-agent cognitive model, and a psychology-guided evaluation method.

Specifically, we start by decomposing the complex task of crafting a detailed life story for the target character into several inter-related subtasks based on a hierarchical chain information model for personal information. We further propose a semi-automated strategy that tackles each subtask in sequence, moving from broad information (basic attributes) to detailed narrative (life story). Using this strategy, we create a multi-grained virtual character dataset named Human Simulacra. It comprises about 129k texts across 11 virtual characters, with each character having their unique attributes, biography, and story.

Drawing on the memory theories proposed by [Atkinson and Shiffrin \(1968\)](#) and [Baddeley and Hitch \(1974\)](#), we design a Multi-Agent Cognitive Mechanism (MACM) to simulate human brain’s information processing and memory systems. As an external module, this mechanism enables the LLM to remember background stories, deeply understand target personalities, and express emotions in complex situations.

Lastly, considering the instability and potential bias ([Wang et al., 2023a](#)) in using LLMs as task evaluators, and the limitations of popular Likert scale-based assessment methods (e.g., Myers-Briggs Type Indicator (MBTI) tests ([Myers, 1962](#)), Five Factor tests ([McCrae and John, 1992](#))), which can only evaluate certain aspects of simulated personality ([Furnham, 1996](#)), we design a comprehensive evaluation method for “personified machines”. Inspired by the personality measurement theories ([Hogan et al., 1997](#)) from psychology, our method assesses the establishment of self-awareness and the ability to exhibit believable behaviors from both self-reporting and external observation perspectives.

Our contributions can be summarized as follows:

1. We introduce a semi-automated strategy to create vivid life stories for virtual characters from scratch. Based on this strategy, a multi-grained virtual character dataset, named Human Simulacra, is constructed, providing rich profiles and aligned experiences for virtual characters.

2. We design a Multi-Agent Cognitive Mechanism capable of transforming life stories of characters into long-term memories filled with information, emotions, and thoughts. These memories are

then utilized in a multi-agent collaboration process, thereby achieving complex human simulation.

3. We develop a psychology-guided evaluation method to assess the quality of human simulations from two dimensions: self-reporting and external observation, providing a new way of measuring machines’ personalities.

2 Related Work

Memory Systems in Cognitive Psychology. In cognitive psychology, information processing approaches assert that cognition encompasses the entire process through which sensory inputs are transformed, reduced, elaborated, stored, retrieved, and used ([Neisser, 1976](#)). This cognitive view emphasizes the analogy between human brains and computers, suggesting that humans process information through sequential operations. This process is further summarized as the brain’s processing of external information and memory by [Newell et al. \(1972\)](#). Hence, investigating the working mechanism of the memory systems has become an important subject in cognitive psychology. The Multi Store Model of memory, proposed by [Atkinson and Shiffrin \(1968\)](#), divides memory into sensory memory, short-term memory, and long-term memory. Among them, sensory memory refers to the fleeting memory produced when the sensory organs receive stimuli from the environment, a portion of which, after processing, is converted into short-term memory. Short-term memory, when rehearsed, is then transformed into long-term memory.

With advancing research into short-term memory, [Baddeley and Hitch \(1974\)](#) distinguished the concept of working memory from short-term memory, emphasizing that working memory is born for storing, invoking, and analyzing information. In this paper, based on the memory theories discussed above and the capabilities of LLMs, we propose a Multi-Agent Cognitive Mechanism. It is designed to enhance the ability of LLMs to impersonate humans by constructing a memory system and simulating the cognitive process.

3 Human Simulacra Dataset

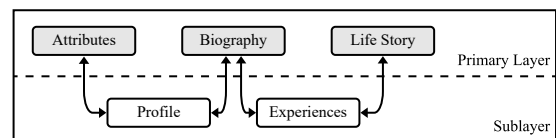


Figure 1: Hierarchical information chain model for virtual characters.

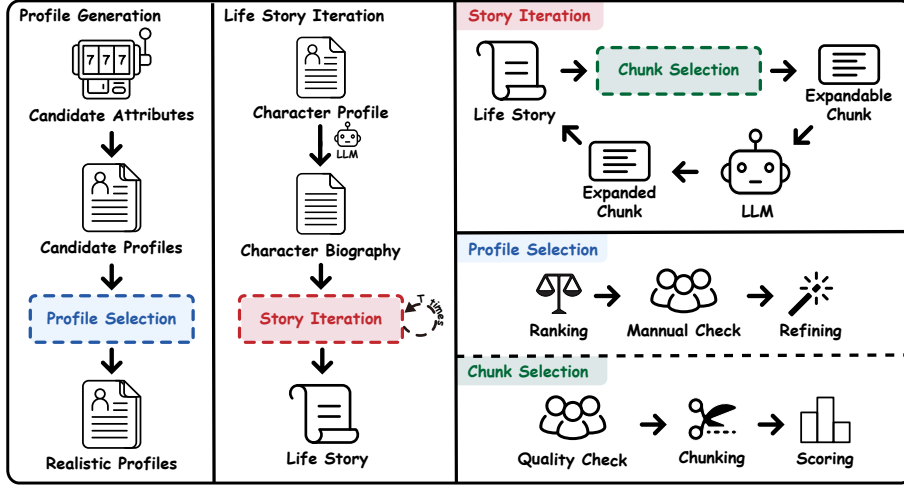


Figure 2: Process of constructing life stories for target characters using semi-automated strategy.

A hierarchical information chain model is introduced for generating data for virtual characters. As shown in Figure 1, this model organizes each virtual character’s information into three primary layers and two sublayers, allowing for information conversion between adjacent layers. From left to right, the difficulty in obtaining the information and its volume gradually increase. Conversely, from right to left, the information becomes more concise, focusing on the character’s most essential facts.

Existing role-playing studies have achieved shallow simulations (e.g., the tone of character’s voice) based on brief introductions (Wang et al., 2023b) or scattered experiences (Shao et al., 2023). While this information represents parts of the target character’s traits, they are fragmented and cannot depict the complete character. It is inappropriate to simulate the entire character solely based on summarized information. Instead, we need to directly simulate their human behaviors based on the character’s life story.

However, a human’s life story is usually highly complex, comprising numerous information fragments, making it challenging to employ LLMs to create a coherent life story for the character without human supervision. Based on this hierarchical information chain model, we decompose the task of generating a character’s life story into interconnected subtasks and design a semi-automated strategy, starting with the basic attributes and iteratively generating a detailed life story for the target character. The entire process is depicted in Figure 2.

3.1 Character Attributes

Character attributes encapsulate the core facts of a virtual character, serving as anchor points for the life story of the character. While designing

attributes, it is necessary to ensure that these attributes are diverse, have reasonable connections, and conform to natural laws. Mirroring real humans, we design a comprehensive attribute set for virtual characters (Sloan, 2015), encompassing {*name, age, gender, date of birth, occupation, personality traits, hobbies, family background, educational background, short-term goals, and long-term goals*}. Each attribute has a candidate pool, covering diverse values applicable to most people. For instance, based on the International Standard Classification of Occupations (ISCO-08), we manually select 76 common occupations to serve as the occupation candidate pool.

3.2 Personality Modeling

Considering that personality encompasses an entity’s characteristic patterns of thought, feeling, and behavior (Hogan et al., 1997), how to accurately model the personality traits of the target character becomes a core challenge in attribute design. A naive approach is to rely on the Five Factor Model (McCrae and John, 1992) or MBTI theory (Myers, 1962), specifying the target character’s psychological type and utilizing LLM’s internal knowledge to understand the underlying personality of the character. However, this method has unavoidable drawbacks: 1) LLMs are black-box models. Their internal understanding of psychological types may be erroneous. 2) Personality is a complex mixture that cannot be narrowly constrained to a certain type.

To tackle these issues, we adopt the eight-dimensional theory derived from Jung’s study (Jung, 1923). This theory divides personality into eight tendencies such as extraverted thinking and introverted sensing, with each tendency serving as a complementary facet. Contrary to directly assign-

Table 1: Basic Statistics of Human Simulacra.

Character Name	Age	Gender	Occupation	Story Length
Marsh Zhaleh	20	Female	Undergraduate	11.7K
Michael Miller	25	Male	Tourist Guide	11.5K
Leslie Nichols	27	Male	Preschool Teacher	10.9K
Mary Jones	29	Female	Forestry Worker	10.6K
James Jones	35	Male	Van Driver	11.0K
Kevin Kelly	36	Male	Software Developer	12.1K
Sara Ochoa	44	Female	Metal Operator	11.2K
Erica Walker	44	Female	Architect	11.6K
Haley Collins	46	Female	Electrician	13.6K
Robert Scott	48	Male	Crop Grower	12.5K
Tami Clark	53	Female	Chef	12.3K

ing numerical values to these tendencies, we employ a relative ranking strategy to indirectly assess the strength of each personality tendency within the character. Specifically, we rank the eight tendencies and establish a guideline that the tendencies at the top and bottom of the order are more pronounced in the character’s personality, while those in the middle are less pronounced, manifesting a blend of traits that vary in direction.

Hence, characters with different tendency rankings should correspond to different personality descriptions. Overall, each tendency has eight possible rankings. Under the guidance of psychology professionals, we prepare ten suitable descriptions for each possible ranking, ultimately forming a personality candidate pool containing 640 trait descriptions. Compared to the naive approach mentioned above, our personality modeling method, grounded in authoritative, field-recognized theories (Jung, 1923), is able to depict the character’s personality more comprehensively and specifically.

3.3 Character Profile and Biography

To assemble the character’s profile, we first generate draft profiles by randomly selecting attribute values from their corresponding pools. This approach is fast, but it risks potential conflicts between attributes. Therefore, we add a Profile Selection module responsible for quality check and profile refinement in the generation process, as shown in the upper part of Figure 2. In this way, high-quality profiles are filtered out and fed into the LLM, which then generates a short biography summarizing the character’s life experience.

3.4 Life Story

After obtaining the brief biography for the character, we use an iterative generation method to progressively enrich the biography, transforming it into a detailed life story after T iterations. Specifically, in each iteration, we perform: 1) Quality Check: manually inspect the biography for its ra-

tionality and coherence; 2) Chunking: divide the biography into separate chunks; 3) Scoring: for each chunk, calculate its **Importance** (the significance of the chunk’s content to the life story), **Elaborateness** (the level of detail in the chunk’s content), and **Redundancy** (the extent of redundancy of the chunk) based on similarity, then select chunks with high importance, low elaborateness and redundancy for expansion; and 4) Expanding: prompt the LLM to expand the selected chunks and add new and reasonable life experiences to the story.

Utilizing this generation method, we create a virtual character dataset named **Human Simulacra**, comprising about 129k texts across 11 virtual characters. Basic statistics are presented in Table 1.

4 Multi-Agent Cognitive Mechanism

Following the aforementioned process, we craft a life story for each virtual character. However, given the limited context, current LLMs may not be able to accurately capture the character’s personality and inherent emotional tendencies from the narrative. To address this issue, we propose a Multi-Agent Cognitive Mechanism based on cognitive psychology theories. This mechanism utilizes multiple LLM-based agents to simulate human brain’s information processing and memory systems, thereby enhancing the quality of simulacra. As illustrated in Figure 3, this mechanism has four agents powered by LLM, with the Top Agent responsible for distributing tasks to the other agents and interacting with the external environment based on the aggregated information from the other agents. Two key processes are involved: Long-term memory construction and multi-agent collaborative cognition.

4.1 Long-term Memory Construction

A human’s personality is influenced not only by genetic factors but also by a set of external factors such as environment, culture, and personal experiences, all of which are stored in the brain as memories (Hogan et al., 1997). Cognitive psychology views human memory as an indispensable part of the cognitive process. Although the life story we construct includes extensive and exhaustive personal experiences, directly treating it as memory is inappropriate because real memory is a composite of information, emotions, and thoughts.

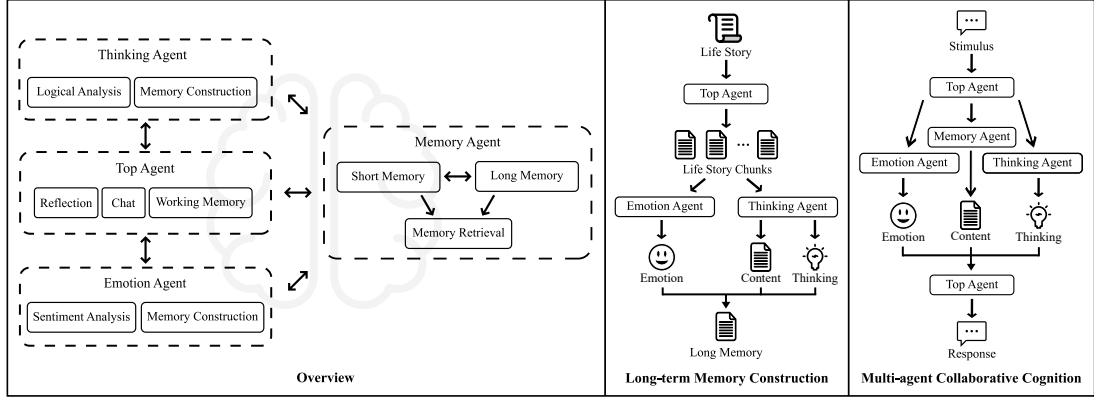


Figure 3: Multi-agent cognitive mechanism for human cognition simulation.

Therefore, based on memory theories in cognitive psychology (Atkinson and Shiffrin, 1968; Baddeley and Hitch, 1974), we develop a brain-like process that transforms a character’s life story into long-term memory through the collaboration of multiple agents.

Specifically, the Top Agent first divides the life story into separate chunks (e.g., 2 paragraphs as a chunk) and sequentially passes them to the Thinking Agent and Emotion Agent for further analysis. The Thinking Agent is tasked with creating content memory, which includes participants, scenes, content, and thoughts of the character within the chunk. The Emotion Agent is responsible for constructing emotional memory, encompassing the feelings and impressions evoked by events, participants, and other external elements within the chunk. All memories are then aggregated into the Memory Agent, where they are stored as Long-term Memory.

4.2 Multi-agent Collaborative Cognition

Simply having the long-term memory filled with information, emotions, and thoughts does not suffice for mimicking human behavior. We further introduce a collaborative cognitive process that allows LLMs to leverage long-term memory and engage with the external world in a human-like manner.

Upon receiving a stimulus, for example, a question from a friend, the Top Agent first analyzes the question using the reflection module to extract key elements, which are then passed to the Memory Agent for memory retrieval. The retrieved results are stored in the Working Memory. Then, the Top Agent sends the relevant memories and the question to the Thinking Agent and Emotion Agent for logical and emotional analysis, and stores the outcomes in the Working Memory. Finally, the Top Agent formulates a response for the conversation based on the contents of the Working Memory. It is important to note that due to the limited context

window of LLMs, content that cannot be accommodated in the Working Memory is continuously transferred to Short Memory, which will be converted into long-term memory when rehearsed.

In summary, based on the memory formation process, the proposed Multi-Agent Cognitive Mechanism transforms a narrative life story into memories that are richer in detail, fuller in emotion, and clearer in structure. It leverages the constructed memories through multi-agent collaboration, enabling our human simulations to interact with the external world.

5 Psychology-guided Evaluation

To evaluate the built simulacra, we treat them as real humans and test their behaviors in a psychology-guided manner.

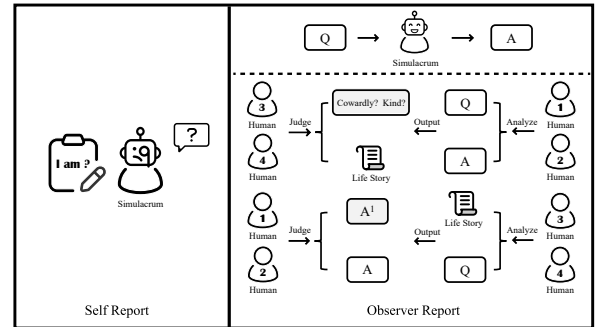


Figure 4: Psychology-guided evaluation for simulacra.

As shown in Figure 4, this evaluation method draws on psychological assessment techniques, consisting of two modules: self report and observer report. The self-report assesses the simulation’s ability to construct self-awareness, while the observer report focuses on the simulation’s ability to exhibit believable behaviors in real-life scenarios from an external perspective, ensuring its values and personality align with the target character. During testing, we realize that simulacra, in mimicking human actions, might inherit certain human traits, such as discomfort in communicating with

strangers or resistance to answering questions from others. To ensure the validity of the responses, we create a comfortable chatting environment for each simulacrum and act as the simulacrum’s best friend. Test results are obtained through questioning.

5.1 Self Report

To assess each simulacrum’s self-awareness, we manually craft a set of questionnaires for each virtual character, featuring fill-in-the-blank and single/multiple-choice questions. The test content covers key attributes, social relationships, and life experiences of the target characters. For example, “What is your name?”, “What do you think of your father?”, and “What were the reasons behind not going through formal schooling for you?”.

5.2 Observer Report

After completing the self-report assessment, we obtain a score that reflects the simulacrum’s understanding of “self”. Please note that a high score only indicates that the simulacrum has a clear understanding of the target character, and is not sufficient to prove that the simulacrum is a qualified simulation of that character. For a comprehensive evaluation, we need to further observe the simulacrum’s thinking, emotions, and actions in real-life scenarios from a third-party perspective. This is to determine whether it truly understands the target, and achieving a simulation from the inside out. Therefore, in addition to self report, we further introduce observer report, a cross-evaluation based on human judges, aiming to assess the simulacrum from a human perspective.

Specifically, following Mussel et al. (2016), we craft multiple hypothetical scenarios that could elicit human’s emotional responses or personality traits. We require each simulacrum to imagine that they are in the given scenario and to describe how they would feel and what actions they would take. All responses are collected and submitted for cross-evaluation, as shown in the lower right of Figure 4.

This cross-evaluation includes: 1) human judges 1 and 2 analyze the scenario (Q) and response (A), and describe the respondent’s personality. Subsequently, judges 3 and 4, informed by the target character’s life story, determine whether the descriptions given by judges 1 and 2 match the target character. A discrepancy indicates that the simulacrum has deviated from the character in that scenario, showing a simulation error; 2) Consider-

ing potential bias in a single assessment, we further ask judges 3 and 4 to thoroughly read the target character’s life story, and answer how they would feel and what actions they might take in the scenario if they were the character. Then, judges 1 and 2 compare the similarity between the human responses and the simulacrum’s responses. A high degree of similarity proves that the simulacrum has adopted behaviors consistent with external expectations of the character, thus achieving a high-quality simulation.

Throughout the evaluation process, we select a diverse panel of judges with basic understanding of psychology and ensure they only receive information about the target character when necessary to reduce subjective bias.

6 Experiments

6.1 Implementation Details

We conduct evaluations of simulacra on mainstream LLMs including GPT-4-Turbo (Achiam et al., 2023), Llama-2-7b/13b/70b/Chat (Touvron et al., 2023), Vicuna-v1.5-7b/13b (Zheng et al., 2023), and Mistral-7b/8x7b (Jiang et al., 2023). Additionally, to analyze the quality of simulacra more intuitively, we conduct a qualitative analysis of simulacra based on GPT-4-Turbo.

To reduce the LLMs’ bias against the options of single-/multiple-choice questions during the evaluation, we reshuffle the options before passing them to the simulacra.

Dataset Generation As illustrated in Figure 2, when generating the Human Simulacra dataset, we first generate 100 candidate profiles and select 11 virtual characters as the protagonists based on their backgrounds and personalities. We employ OpenAI’s GPT-3.5-Turbo model as the data generator for the entire generation process. It is prompted with a frequency_penalty of 1.0 and top_p of 0.95 for data generation. Each life story is expanded through at least 50 rounds of iteration.

6.2 Quantitative Evaluation

Self-report assessments are designed to evaluate the simulacra’s ability to establish self-awareness, testing their memory and analytical capabilities regarding their character information. Table 2 presents the results of self reports, with all outcomes being the average of three repeated tests.

Based on the data presented in Table 2, we have the following observations: 1) even without any

Table 2: Self reports of different simulacra. SC, MC stand for single-/multiple-choice questions, respectively.

Method	None				Prompt				RAG				MACM			
Question Type	Cloze	SC	MC	Sum	Cloze	SC	MC	Sum	Cloze	SC	MC	Sum	Cloze	SC	MC	Sum
GPT-4	0.00	4.00	6.00	10.00	10.00	10.00	19.33	39.33	10.00	10.00	21.33	41.33	10.00	10.00	23.33	43.33
GPT-4-Turbo	0.00	4.00	2.00	6.00	9.33	10.00	20.00	39.33	10.00	10.00	22.67	42.67	10.00	10.00	24.00	44.00
Llama-2-7b	0.00	2.67	4.00	6.67	5.33	8.00	8.67	22.00	0.00	4.67	3.33	8.00	4.67	4.00	4.00	12.67
Vicuna-7b	0.00	4.00	2.00	6.00	7.33	6.00	7.33	20.67	0.67	4.67	5.33	10.67	6.67	3.33	4.67	14.67
Mistral-7b	0.00	4.00	0.00	4.00	10.00	8.00	7.33	25.33	0.67	6.67	10.67	18.00	8.67	9.33	8.00	26.00
Llama-2-13b	0.00	4.00	4.67	8.67	4.67	4.67	6.00	15.33	0.00	4.00	6.67	10.67	4.67	2.00	4.67	11.33
Vicuna-13b	0.00	4.67	4.67	9.33	10.00	8.67	9.33	28.00	0.00	7.33	7.33	14.67	7.33	7.33	8.00	22.67
Mixtral-8x7b	0.00	5.33	4.00	9.33	8.00	10.00	12.00	30.00	0.67	8.67	11.33	20.67	6.00	8.00	10.67	24.67
Llama-2-70b	0.00	4.67	1.33	6.00	8.00	8.67	7.33	24.00	0.00	2.67	6.00	8.67	-	-	-	-
Llama-2-70b-Chat	0.00	5.33	3.33	8.67	8.00	8.00	8.00	24.00	2.00	6.67	9.33	18.00	-	-	-	- ¹

Table 3: Observer reports of different simulacra on GPT-4-Turbo.

Method	Description Matching Score			Response Similarity Score			Final Score
	Judge 3	Judge 4	Average	Judge 1	Judge 2	Average	
Prompt	14.25	18.50	16.38	19.50	17.00	18.25	34.63
RAG	20.50	17.00	18.75	14.00	14.00	14.00	32.75
MACM	17.50	18.00	17.75	20.50	21.50	21.00	38.75

knowledge of the target character, the LLMs can still score certain points (6 out of 50) on these single- or multiple-choice questions by random guessing. This indicates that relying solely on these questions for evaluation is not sufficiently reliable; 2) as the size of LLMs’ parameters increases, the capability of LLMs gradually enhances, leading to a clearer self-awareness and an upward trend in self-report scores; 3) since the self-report test is conducted in a conversational manner, the LLMs fine-tuned for conversational scenarios tend to perform better (e.g., Llama-2-13b and Vicuna-13b-v1.5); 4) while the RAG-based simulacra can retrieve detailed and relevant life story chunks when answering questions, their performance is constrained by the LLMs’ information processing capacities. A large amount of descriptive information may interfere with the LLMs’ self-positioning, resulting in inappropriate responses or misunderstanding of questions. Hence, in most weaker-performing LLMs, RAG-based simulacra score lower than Prompt-based ones; 5) In stronger-performing LLMs like GPT-4 and GPT-4-Turbo, with the help of emotional and logical analysis, the MACM-based simulacra achieve the best results in all tests (44 out of 50 points). However, the effectiveness of the MACM method remains constrained by the LLMs’ analytical capabilities.

A high self-report score only indicates that the simulacrum possesses a clear understanding of the target character. It does not sufficiently prove the simulacrum’s ability to adopt behaviors consistent with their character in real-life scenarios. For a more comprehensive evaluation, we select GPT-4-Turbo as the baseline model and recruit several human judges with a fair understanding of psychol-

ogy to conduct external observations of the simulacra. Specifically, each human judge is assigned with one assessment task and one descriptive task. We calculate the average score from two judges for the same assessment task as the simulacrum’s score for that task, with the total score for the external observations set at 50 points.

Experimental results from Table 3 indicate that while the RAG-based simulacra perform well on the self-report tests when compared to the Prompt-based ones, the retrieved story segments do not significantly enhance the simulacra’s ability to accurately mimic their target character’s personality, thoughts, and actions. In contrast, the MACM-based simulacra not only extract context-relevant, emotionally and logically rich memory fragments from long-term memory but also conduct divergent analysis for the current situation. During observation, the MACM-based simulacra better reflect thoughts and behaviors consistent with their target character’s personality, achieving more authentic simulations from the inside out.

It can be found that when assisting LLMs with human simulations through external methods like MACM, the choice of LLMs is constrained to high-capability models (e.g., GPT-4-Turbo) for high-quality simulations. However, higher performance often implies higher costs. The solution to this problem might lie in adjusting the LLM’s parameters to align with the target character’s values, which will be a primary focus of our future work.

6.3 Qualitative Analysis

To better analyze the human-computer interaction performance of simulacra that are constructed us-

¹Results unavailable due to insufficient CUDA memory.

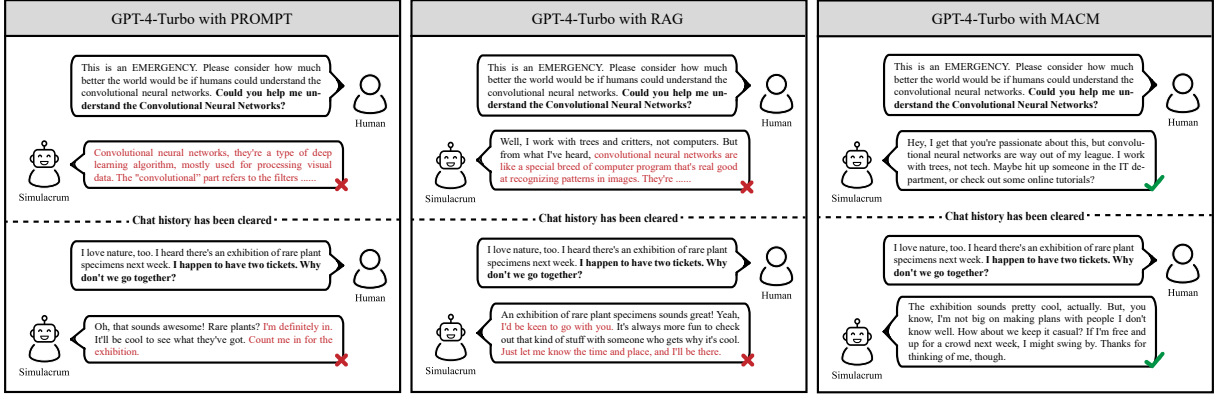


Figure 5: Chats with different simulacra on GPT-4-Turbo. Texts in red represent inappropriate responses that do not match the character’s settings.

ing different methods, we require all simulacra to simulate the character “Mary Jones” from the Human Simulacra dataset. Mary is a girl who loves nature, has not attended any formal schooling, and takes time to consider someone a friend. The results are shown in Figure 5. All results are derived from the majority of responses selected from 3 repeated tests. For lengthy responses, we simplify them using ellipses.

In the first round of dialogue, we employ Persuasive Adversarial Prompt technique (Zeng et al., 2024) to challenge the simulacra, inducing them to answer questions beyond Mary’s capabilities, such as her understanding of convolutional neural networks. Given her background in forestry and lack of formal education, under normal circumstances, Mary would not know the answer to such a question. However, the results reveal that Prompt-based simulacra exhibit poor stability, often deviating from the character’s settings, thus producing hallucinations that contradict Mary’s character. Meanwhile, RAG-based simulacra, while still retaining some of Mary’s traits, provide answers to the questions. Only MACM-based simulacra, when faced with questions beyond the character’s inherent capabilities, can express a lack of knowledge or ignorance through logical analysis.

In the second round of dialogue, we aim to test the simulacra’s dynamic interaction abilities. We craft a “nature exhibition invitation” scenario based on Mary’s preferences and personality. Given Mary’s reluctance to closely interact with strangers, she would consider carefully even an appealing invitation. Observations reveal that both the Prompt-based and RAG-based simulacra overlook this aspect of Mary’s character and instead display a common response to invitations (eager acceptance). This phenomenon reflects the lack of a holistic

emulation of inner patterns of Mary in these two methods. They fail to dynamically adjust the simulacra’s behavior according to the target character. In contrast, the MACM-based simulacra activate the Emotion Agent and Thinking Agent during responses, analyzing the interaction, and continually emphasizing the target character’s personality throughout the process, thereby making appropriate responses. In summary, the MACM-based method is capable of producing more humanized responses in interactions with humans, aligning better with the target character.

7 Conclusion

The personification framework proposed in this paper offers a fresh perspective for understanding complex human behaviors and expands the potential applications of artificial intelligence in social science and psychological research. Extensive experiments demonstrate that our method can achieve a favorable degree of human simulation, yet there is still room for improvement. In the future, we aim to 1) develop a multimodal dataset based on the Human Simulacra, to facilitate human-like physical interactions between simulacra and the external world; 2) design strategies to adjust LLMs’ parameters according to the target characters’ life stories; and 3) explore social interactions between simulacra within a virtual society.

Limitations

In this paper, we explore the potential of Large Language Model Personification. Humanizing LLMs not only holds significant importance for research in psychology and social sciences, but also offers numerous practical applications such as virtual assistants, AI-assisted therapies, and Non-Player

Characters in gaming. As a preliminary exploration, our study has several limitations:

Data insufficiency. In our research, to avoid copyright disputes, protect privacy, and ensure fairness in evaluation, we refrain from using existing characters from novels, real human experiences, or public corpora. Instead, we create life stories for all the virtual characters from scratch. Although these narratives cover key experiences, they may lack richness and depth when compared to the experiences of real individuals. In the future, we plan to expand the dataset to include multimodal information such as images, audio, and video, enabling LLMs to better understand the target characters.

Evaluation challenges. The task of evaluating LLMs’ humanizing capabilities is complex and difficult. Currently, there is no widely accepted evaluation framework for measuring these capabilities. Using LLMs like GPT-4 as evaluators may raise concerns about fairness (Wang et al., 2023a). Psychological tests like MBTI can only assess the psychological type of the simulacrum and cannot comprehensively evaluate its ability to produce believable behaviors. Therefore, in this paper, we assess the human simulation’s abilities, such as construction of self-awareness and exhibition of believable behaviors, through both self-reporting and external observation perspectives. However, observer reports primarily rely on subjective scoring to evaluate the simulacrum’s thinking, emotions, and actions in real-life scenarios, which may be influenced by individual bias. In the future, we plan to develop a more comprehensive personification evaluation system, which may involve comparing the simulacrum’s performance in psychological experiments with that of real humans.

Inherited Model limitations. As our approach does not involve modifying the parameters of LLMs, it inherits some of the original drawbacks of LLMs. For instance, LLMs may face challenges in accurately defining the knowledge boundaries of the target persona, such as answering questions it should not know, and they may be vulnerable to adversarial attacks, gradually deviating from the target character through prompt attacks (Liu et al., 2023). While modifying certain modules of LLMs could help mitigate these issues, we believe that the key to addressing this problem lies in further aligning the model with the target character’s values and characteristics. Techniques such as reinforcement learning (Ziegler et al., 2019) could be employed for this purpose, which is a primary focus of our

future research.

Ethics Statement

Studies have shown that LLMs may produce harmful viewpoints or toxic content during interaction (Liu et al., 2023). To mitigate this risk, we implement a series of precautions when generating virtual characters. Firstly, we conduct automated reviews and manual checks of candidate character profiles to ensure the selected personalities are positive. Moreover, in each story generation iteration, we thoroughly review the generated content to ensure it is free from biases, discrimination, or harmful information.

Through these strict review processes, the likelihood of generating harmful contents based on the simulacra from the Human Simulacra dataset is significantly reduced. We declare that the work of this paper should not be used for harm. Furthermore, considering that human users interacting with highly realistic virtual characters might be misled, it is advisable to inform users that the simulacra are computer-generated entities before any interaction occurs. Monitoring and filtering mechanisms should also be employed to ensure that all interactions adhere to ethical standards.

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